ENN595-1 Project Summary Oral Presentation: Master Script

Project Title: Investigation of Primitive-Space Scene Change Detection in Underwater 3D Gaussian Splatting Models
Presenter: Joshua Hecke (n11585382)
School / Faculty: School of Electrical Engineering & Robotics | QUT Centre for Robotics
Supervisory Team: Prof. Niko Sünderhauf (Principal), Prof. Will Browne, Prof. Matthew Dunbabin
Public Code Repository: https://github.com/Heckus/UMP
Strict Presentation Time Limit: 8 Minutes Maximum (Script timed to **7m 45s** with a 15-second safety buffer) + **2 Minutes Q&A**
Total Target Spoken Words: ~1,030 words (Delivered at a calm, authoritative **130–135 words per minute**)

Quick Reference: Presenter's 8-Minute Pacing Dashboard

Keep this summary table visible on your podium or folder for instant time checks during the talk.

Slide #	Slide Title	Visual Asset / Screen Action	Clock Time	Duration	Words	Pace Check
1	Title & Core Research Question	Hexagon title frame & public repo badge	`0:00 – 0:30`	`30s`	68	Authoritative
2	Ecological Context & 2D Bottleneck	3-column reef problem & 45% rugosity card	`0:30 – 1:05`	`35s`	78	Urgent / Clear
3	Concept: 3D Gaussian Splatting (2023)	Kerbl et al. paper & bicycle 3DGS loop	`1:05 – 1:35`	`30s`	67	Explanatory
4	Concept: Primitive-Space SCD (2026)	GS-Diff paper & Decoupled Kernels	`1:35 – 2:10`	`35s`	78	Methodical
5	Concept: Underwater Optical 3DGS (2025)	Jaffe-McGlamery equation & 6 models	`2:10 – 2:45`	`35s`	78	Technical
6	Concept: SuGaR Surface Regularization	Guédon paper & garden mesh GIF	`2:45 – 3:20`	`35s`	79	Problem-Solving

Slide #	Slide Title	Visual Asset / Screen Action	Clock Time	Duration	Words	Pace Check
7	Proposed 6-Stage CONOPS Architecture	High-res CONOPS map + HPC stages	3:20 – 4:00	40s	89	**Midpoint (4:00)**
8	Strategy: Why Submerged3D (20 Img × 4 Sites)	4-site grid GIF + Engineering buffer	4:00 – 4:40	40s	88	Justifying
9	Production Target: AIMS EcoRRAP (1.35 TB)	EcoRRAP coral transect matrix	4:40 – 5:10	30s	68	Visionary
10	Progress: Aqua HPC & 15/15 Milestone	11 Conda envs & 15/15 success table	5:10 – 5:45	35s	78	Rigorous
11	Progress: Quantitative Benchmark Results	Kwaj & Tokai metrics table + Orbit GIFs	5:45 – 6:30	45s	99	Analytical
12	Progress: Per-Model SuGaR 3D Meshes	Static bright mesh renders (Kwaj & Tokai)	6:30 – 7:00	30s	68	Decisive
13	Progress: Primitive-Space O-SCD Results	4-panel change mask + Orbiting ghost can	7:00 – 7:35	35s	78	Climax / Proof
14	Priorities for Next Steps & Roadmap	Phase 2–4 Gantt roadmap + Thank you	7:35 – 8:00	25s	56	Forward-Looking
Q&A	**Examiner Defense Session**	**Switch to slide 11, 10, or 13 as needed**	**8:00 – 10:00**	**2m 00s**	—	Calm & Precise

Printable Slide-by-Slide Script Cards

Slide 1: Title Slide & Core Research Question

Timing: `0:00 – 0:30` (30s) | **Word Count:** 68 words | **Pacing:** Calm, confident, formal
Rubric Focus: *Criterion 1 (Context & Well-Formulated Research Question)*
Visual On Screen: Title slide with dark-navy theme, supervisor names, and public repository callout (`github.com/Heckus/UMP`).

Stage Directions & Delivery:

- **[STAND TALL, EYE CONTACT ACROSS PANEL]**
- *[Pronounce: Hecke = HECK-ee | Sünderhauf = SOON-der-howf]*

> "Good morning everyone. My name is **Joshua Hecke**, and under the supervision of Professor Niko Sünderhauf, Professor Will Browne, and Professor Matthew Dunbabin at the QUT Centre for Robotics, my research investigates **Primitive-Space Scene Change Detection in Underwater 3D Gaussian Splatting Models**.
>
> Our core research question asks: **How can we detect fine-scale structural growth, breakage, and bleaching directly inside 3D Gaussian primitive space, without being deceived by fluctuating underwater turbidity and backscatter?**"

Transition to Slide 2: **[ADVANCE SLIDE]** "To understand why this is such an urgent challenge, let us look at how coral reefs are monitored today."

Slide 2: Project Context: Why 3D Underwater Change Detection?

Timing: `0:30 – 1:05` (35s) | **Word Count:** 78 words | **Pacing:** Engaging, problem-focused
Rubric Focus: *Criterion 1 (Clear, Critical Analysis of Ecological Context & Limitations)*
Visual On Screen: 3-column analysis: (1) Ecological Crisis, (2) 2D Diver Survey Limits, (3) Marine Photogrammetry Challenge.

Stage Directions & Delivery:

- **[POINT TO: 45% Rugosity Stat in Column 2]**
- *[Pronounce: rugosity = roo-GOSS-ih-tee | photogrammetry = foe-toe-GRAM-eh-tree]*

> "Coral ecosystems like the Great Barrier Reef face unprecedented thermal bleaching and storm destruction. Ecological programs like AIMS EcoRRAP need to monitor **millimetre-to-centimetre changes** over time.
>
> However, standard monitoring still relies on two-dimensional planar photo-transects. Recent studies show 2D surveys underestimate structural reef rugosity by **up to 45%**, completely missing complex vertical branching and under-canopy health.
>
> Yet moving to 3D underwater vision is notoriously difficult: between survey passes, changing light, water haze, and floating particles corrupt standard image comparison."

Transition to Slide 3: **[ADVANCE SLIDE]** "To overcome this, our foundation builds upon 3D Gaussian Splatting."

Slide 3: Understanding the Concept (2023): 3D Gaussian Splatting

Timing: `1:05 – 1:35` (`30s`) | **Word Count:** 67 words | **Pacing:** Clear, instructional, explanatory
Rubric Focus: *Criterion 1 (Mastery of Foundational Technologies & Formulations)*
Visual On Screen: Kerbl et al. (SIGGRAPH 2023) paper highlight and real-time bicycle scene rendering loop.

Stage Directions & Delivery:

- **[GESTURE TO THE EXPLICIT PRIMITIVES CALLOUT]**
- *[Pronounce: anisotropic = an-eye-so-TROP-ik | covariance = co-VAIR-ee-ance]*

> "Introduced by Kerbl et al. in 2023, 3D Gaussian Splatting revolutionised computer vision.
>
> Unlike Neural Radiance Fields—which bury scene geometry inside slow, black-box neural networks—3DGS represents the world **explicitly** as millions of 3D Gaussian ellipsoids. Each primitive carries an exact 3D position, anisotropic covariance shape, opacity, and spherical harmonic colour.
>
> This delivers real-time rendering at over thirty frames per second, while giving us direct mathematical access to the physical primitives themselves."

Transition to Slide 4: **[ADVANCE SLIDE]** "In 2026, QUT researchers leveraged these explicit primitives to reinvent change detection."

Slide 4: Understanding the Concept (2026): Primitive-Space Change Detection

Timing: `1:35 – 2:10` (35s) | **Word Count:** 78 words | **Pacing:** Technical, authoritative
Rubric Focus: *Criterion 1 & 2 (Cutting-Edge Concept & Decoupled Change Formulations)*
Visual On Screen: Galappaththige et al. (QUT 2026) paper banner, Decoupled Kernel cards, and `concept_oscd_paper.gif`.

Stage Directions & Delivery:

- **[POINT TO: Green Geometric Kernel, then Orange Appearance Kernel]**
- *[Pronounce: Galappaththige = gah-lahp-PAHTH-tih-gay | SuGaR = SHOO-gahr]*

> "Galappaththige et al. introduced **GS-Diff** and **O-SCD**, operating directly on 3D primitives rather than comparing rendered 2D images.
>
> Crucially, this architecture decouples change into two independent mathematical kernels:
>
> - A **Geometric Kernel**, which detects true physical additions or structural breakage;
> - And an **Appearance Kernel**, which isolates surface colour shifts like thermal coral bleaching.
>
> Importantly, the authors concluded that unconstrained Gaussians can drift over time, explicitly recommending surface regularization like SuGaR to anchor geometry."

Transition to Slide 5: **[ADVANCE SLIDE]** "However, neither 3DGS nor O-SCD were designed to function through water."

Slide 5: Extending the Concept (2025): Optical Water Restoration

Timing: `2:10 – 2:45` (35s) | **Word Count:** 78 words | **Pacing:** Confident, scientific
Rubric Focus: *Criterion 1 & 2 (Physics-Grounded Marine Domain Adaptation)*
Visual On Screen: Jaffe-McGlamery optical formation equation and animated comparisons (`orbit_3d_uir_kwaj.gif`, `orbit_seasplat_kwaj.gif`).

Stage Directions & Delivery:

- **[POINT TO: Equation Box — Backscatter vs. Direct Transmission]**
- *[Pronounce: Jaffe-McGlamery = JAFF-ee mc-GLAM-er-ee | albedo = al-BEE-doh]*

> "Standard 3DGS treats water haze as physical geometry, creating dense swarms of floating ghost splats.
>
> Recent underwater architectures embed the **Jaffe-McGlamery optical formation model** directly into the differentiable rasterizer. This explicitly decouples the medium's wavelength-dependent attenuation and backscatter from true benthic scene radiance.
>
> Across our benchmark, we evaluated six distinct architectures across physics-based methods like **SeaSplat** and **3D-UIR**, alongside restoration models like **Gaussian Splashing** and **WaterSplatting**, isolating true underwater colour and crisp geometry."

Transition to Slide 6: **[ADVANCE SLIDE]** "To clean up the remaining water-column noise, we introduce surface regularization."

Slide 6: Extending the Concept (2024): SuGaR Surface Regularization

Timing: 2:45 – 3:20 (35s) | **Word Count:** 79 words | **Pacing:** Problem-solving, precise
Rubric Focus: *Criterion 2 (Creative Solution to Water Floaters & Surface Alignment)*
Visual On Screen: Guédon & Lepetit (CVPR 2024) paper, SDF density formula, and garden mesh GIF.

Stage Directions & Delivery:

- **[EMPHASIZE: 85–92% Floater Removal]**
- *[Pronounce: Guédon = geh-DOHN | Poisson = PWAH-sohn]*

> "Following the recommendation from the O-SCD authors, our pipeline integrates **SuGaR** by Guédon and Lepetit into Stage 5 for two vital reasons.
>
> First, its Signed Distance Field normal-consistency prior forces ellipsoids flat against actual surfaces, **pruning 85 to 92 percent of floating water-column artifacts**.
>
> Second, it reconstructs a watertight, UV-textured Poisson surface mesh for every model. This gives marine scientists millimeter-level surface geometry and physical area measurements, preventing unconstrained Gaussian drift between surveys."

Transition to Slide 7: **[ADVANCE SLIDE]** "Here is how all these components unify into our operational architecture."

Slide 7: Proposed Solution: End-to-End 6-Stage CONOPS

Timing: `3:20 – 4:00` (`40s`) | **Word Count:** 89 words | **Pacing:** Structured, authoritative
Rubric Focus: *Criterion 2 (Clear Methodology, Engineering Rigour, Public Reproducibility)*
Visual On Screen: Full CONOPS pipeline map (`conops.jpg`), 6-stage HPC workflow badges, and public repository reference box.

Stage Directions & Delivery:

- **[POINT TO: Public GitHub Callout at top: `github.com/Heckus/UMP`]**
- **[SWEEP HAND: Left to Right across Stages 1 through 6]**

> "This CONOPS illustrates our complete temporal change detection pipeline.
>
> Multi-temporal reef passes stream through:
>
> 1. Automated data validation;
> 2. Structure-from-Motion in COLMAP;
> 3. Monocular depth inversion via Depth-Anything-V2;
> 4. Underwater optical 3DGS training;
> 5. SuGaR surface regularization and mesh extraction; and
> 6. Held-out quantitative evaluation.
>
> All custom PBS batch scripts, CUDA rasterizer patches, and automation code referenced throughout this presentation are published on GitHub at **github.com/Heckus/UMP**.
>
> I engineered this into a turnkey six-stage automated pipeline operating on the **QUT Aqua supercomputer**."

Transition to Slide 8: **[ADVANCE SLIDE]** "Before scaling to terabytes of coral, we needed a rigorous engineering test suite."

Slide 8: Experimental Strategy: Why the Submerged3D Test Dataset?

Timing: `4:00 – 4:40` (`40s`) | **Word Count:** 88 words | **Pacing:** Strategic, justified, self-aware
Rubric Focus: *Criterion 2 (Experimental Strategy, Engineering Buffer, Transparent Limitations)*
Visual On Screen: 20-image × 4-location animated grid (`submerged3d_4locations_20images.gif`) and 3-column justification.

Stage Directions & Delivery:

- **[EMPHASIZE: Proactive engineering buffer while DTA is processed]**
- **[POINT TO: 20-image grid cycling through the 4 shipwreck sites]**

> "Why begin with the Submerged3D dataset—specifically twenty images across four shipwreck sites?
>
> First, it acted as a **proactive engineering buffer**. While our primary 1.35-terabyte coral dataset undergoes AIMS legal Data Transfer Agreement review, this compact benchmark allowed rapid development. Twenty images kept full six-stage training to under seventy minutes per run, allowing ten complete debugging cycles across eleven Conda environments.
>
> Second, sparse views stress-tested our pipeline: low-contrast imagery on Cormoran and Isro exposed feature-matching limits early, proving our pre-flight data guards work."

Transition to Slide 9: **[ADVANCE SLIDE]** "This brings us to our primary ecological target."

Slide 9: Production Target: Why the AIMS EcoRRAP Dataset (1.35 TB)?

Timing: `4:40 – 5:10` (30s) | **Word Count:** 68 words | **Pacing:** Visionary, impactful
Rubric Focus: *Criterion 1 & 2 (Ecological Significance & Phase 2 Staging)*
Visual On Screen: EcoRRAP 3-column card analysis and Phase 1 vs. Phase 2–4 comparison matrix.

Stage Directions & Delivery:

- **[POINT TO: High-Density 70% Overlap Metric]**
- **[CONFIDENT TONE: Pipeline is ready and waiting for data]**

> "Our production target is the **1.35-terabyte AIMS EcoRRAP repository**.
>
> Unlike sparse test sets, EcoRRAP features dense, seventy-percent-overlap stereoscopic photogrammetry capturing permanent Great Barrier Reef quadrats across seasonal bleaching and cyclone events.
>
> Our pipeline will allow marine ecologists to walk through interactive 3D digital twins, automatically isolating bleaching colour shifts from physical colony growth or breakage. Our turnkey HPC scripts are fully staged for immediate ingestion upon legal clearance."

Transition to Slide 10: **[ADVANCE SLIDE]** "Now let us examine the concrete engineering milestones achieved this semester."

Slide 10: Progress: QUT Aqua HPC & The 15 / 15 Benchmark Milestone

Timing: `5:10 – 5:45` (35s) | **Word Count:** 78 words | **Pacing:** Energetic, rigorous, triumphant
Rubric Focus: *Criterion 2 (Outstanding Engineering Progress & Forensic Technical Solutions)*
Visual On Screen: KPI metric row (15/15 Pass, 11 Conda Envs, A100/H100 GPUs), Technical Fixes list, and clean green 15/15 Execution Table.

Stage Directions & Delivery:

- **[GESTURE PROUDLY TO: Green "PASS (2/2)" Column in Table]**
- *[Pronounce: CUDA = KOO-duh | ABI = A-B-I | RANSAC = RAN-sak]*

> "Open-source underwater 3DGS codebases are notorious for conflicting dependencies, spanning Python 3.7 to 3.12 and incompatible CUDA rasterizers.
>
> I engineered an automated provisioning suite on QUT Aqua, isolating **eleven Conda environments** across NVIDIA A100 and H100 nodes. Across ten intensive audit cycles, we diagnosed and resolved critical bugs—including COLMAP library linking, custom CUDA rasterizer gradient shapes, and SuGaR numerical NaN collapses.
>
> This delivered our primary milestone: **15 out of 15 full-lifecycle benchmark runs succeeded with Exit Code 0**."

Transition to Slide 11: **[ADVANCE SLIDE]** "Let us inspect the quantitative performance of these models."

Slide 11: Progress: Quantitative & Visual Underwater 3DGS Benchmark

Timing: `5:45 – 6:30` (45s) | **Word Count:** 99 words | **Pacing:** Analytical, deliberate, metric-focused
Rubric Focus: *Criterion 2 & 3 (Technically Sound Evaluation, Media Polish, Rigorous Trade-off Analysis)*
Visual On Screen: Master Quantitative Table (`Kwajalein` & `Tokai Maru`), and 4 orbiting model GIFs (`3D-UIR`, `SeaSplat`, `Gaussian Splashing`, `WaterSplatting`).

Stage Directions & Delivery:

- **[POINT TO: 3D-UIR SSIM in Table: 0.8739 and 0.7085]**
- **[POINT TO: SeaSplat Albedo Recovery in Tokai Orbit GIF]**

> "Here are our quantitative and visual results across Kwajalein and Tokai Maru.
>
> While Gaussian Splashing achieved the highest raw PSNR at 27.93 dB on Kwajalein, its structural similarity collapsed under the heavy blue turbidity of Tokai Maru to 0.6142 due to severe over-smoothing. WaterSplatting was our fastest architecture at 43 minutes with the lowest perceptual error.
>
> However, for change detection, structural integrity is paramount: **3D-UIR achieved the highest structural similarity on both scenes**—0.8739 and 0.7085—preserving fine metallic railings and biological textures. Combined with SeaSplat's superior physical backscatter removal, these two models form our winning backbones."

Transition to Slide 12: **[ADVANCE SLIDE]** "Next, we examine the extracted 3D surface meshes from Stage 5."

Slide 12: Progress: Per-Model SuGaR 3D Surface Meshes

Timing: `6:30 – 7:00` (`30s`) | **Word Count:** 68 words | **Pacing:** Decisive, impactful
Rubric Focus: *Criterion 2 (Creative Engineering Discovery & Mesh Metrology Proof)*
Visual On Screen: Static bright renders of Kwajalein and Tokai Maru meshes (`sugar_kwaj_mesh_bright.png`, `sugar_tokai_mesh_bright.png`) and `--output_dir_tag` isolation badge.

Stage Directions & Delivery:

- **[POINT TO: High-resolution mesh renders showing crisp hull surfaces without floaters]**
- **[MENTION: Public code fix on GitHub]**

> "In Stage 5, every model generates a one-million-face watertight mesh.
>
> During inspection, we uncovered a critical flaw in vanilla SuGaR: it cached meshes by scene name alone, causing all models to overwrite the first model's geometry. We engineered per-model namespace isolation via `--output_dir_tag`, published in our repository.
>
> As seen in these static renders of Kwajalein and Tokai Maru, SuGaR successfully eliminated water floaters and recovered clean, measurable 3D hulls."

Transition to Slide 13: **[ADVANCE SLIDE]** "Finally, we validated our primary objective: Primitive-Space Change Detection."

Slide 13: Progress: Primitive-Space Scene Change Detection (O-SCD)

Timing: `7:00 – 7:35` (35s) | **Word Count:** 78 words | **Pacing:** Climax, authoritative, highly engaged
Rubric Focus: *Criterion 2 & 3 (Demonstration of Primary Thesis Goal & Sophisticated Visual Proof)*
Visual On Screen: 4-panel multi-view change mask animation + 3D Orbiting Turbo Change Heatmap showing the thermal ghost of the removed red can.

Stage Directions & Delivery:

- **[POINT TO: Volumetric thermal "ghost" in the orbiting 3D heatmap]**
- **[EMPHASIZE: Zero false positives from camera angle changes]**

> "Here is our primary change detection engine—O-SCD—operating end-to-end on our dual-state benchmark.
>
> On the left, across 122 viewpoints, the network accurately segments added, moved, and removed objects without false positives caused by camera movement.
>
> On the right, we synthesized direct **3D primitive-space thermal heatmaps**. Notice how the 3D splat captures the volumetric thermal 'ghost' of the removed red soda can, while the updated scene achieves over 27 dB PSNR and 0.83 SSIM."

Transition to Slide 14: **[ADVANCE SLIDE]** "This brings us to our immediate research priorities and roadmap."

Slide 14: Priorities for Next Steps: Phase 2–4 EcoRRAP Roadmap

Timing: `7:35 – 8:00` (25s) | **Word Count:** 56 words | **Pacing:** Crisp, forward-looking, definitive finish
Rubric Focus: *Criterion 2 (Clear, Actionable Future Plan & Confident Conclusion)*
Visual On Screen: Unified Phase 3 Architecture diagram, Phase 2–4 roadmap milestones, and professional acknowledgment.

Stage Directions & Delivery:

- **[STEP FORWARD, CONFIDENT POSTURE]**
- **[SMILE, OPEN HANDS TOWARD THE PANEL]**
- **[FINISH PROMPTLY AT ~7m 50s — EXACTLY ON TIME!]**

> "Looking ahead: In Phase 2, as the AIMS agreement clears, our pipeline will ingest the 1.35-terabyte EcoRRAP dataset.
>
> In Phase 3, we will graft our winning 3D-UIR and SeaSplat optical backbones directly into the O-SCD change engine.
>
> This will deliver an automated, turbidity-invariant 3D monitoring tool for coral reef conservation.
>
> Thank you, and I look forward to your questions."

2-Minute Q&A Defense Battle Card (Criterion 3: 8 pts)

*Deliver crisp, authoritative answers using the **PREP** method: **P**oint (5s) → **R**eason (10s) → **E**vidence (10s) → **P**oint (5s). Keep each answer strictly between 25 and 35 seconds.*

Question 1: "Why only evaluate Kwajalein and Tokai Maru in the full benchmark table if Submerged3D contains 4 shipwreck locations?"

- **Target Response Time:** 30 seconds
- **Key Evidence:** COLMAP convergence on 20 sparse views; diverse optical regimes.

> **Response:**
> "All four locations were tested in Stage 2 Structure-from-Motion. Because Submerged3D provides only twenty sparse frames per site, the extreme low-contrast green haze on *Cormoran* and *Isle Royale* prevented COLMAP SIFT features from converging into a single dense model across that few views.
> That was an essential experimental finding—it validated our automated pre-flight checks and proved dense coverage is essential. *Kwajalein* and *Tokai Maru* converged completely, giving us two ideal, contrasting optical regimes: medium backscatter with biofouling versus extreme deep-blue wavelength attenuation, perfectly testing all seven 3DGS models."

Question 2: "How do you prevent water turbidity differences between Time 0 and Time 1 from causing false positives in primitive-space change detection?"

- **Target Response Time:** 30 seconds
- **Key Evidence:** Three-tier defense: Jaffe-McGlamery optical model + SuGaR SDF pruning + Dual-kernel differencing.

> **Response:**
> "We solve this through a three-stage architectural defense:
> First, our optical backbones—3D-UIR and SeaSplat—embed the Jaffe-McGlamery equation to explicitly decouple water attenuation and backscatter during training, recovering true benthic albedo.
> Second, SuGaR applies a Signed Distance Field normal prior that prunes over 85% of floating water-column Gaussians before differencing begins.
> Finally, O-SCD compares primitives in 3D rather than 2D pixels, using independent geometric and appearance kernels so residual turbidity shifts cannot mimic physical structural displacement."

Question 3: "Why extract SuGaR 3D surface meshes if your primary goal is detecting change directly on Gaussian primitives?"

- **Target Response Time:** 28 seconds
- **Key Evidence:** Literature recommendation (Galappaththige et al., 2026); metric surface evaluation; floater pruning.

> **Response:**
> "There are two major reasons:

> First, the authors of the 2026 O-SCD paper explicitly identified that unconstrained Gaussians drift across surfaces between independent runs, recommending SuGaR surface regularization to bind Gaussians to physical reality.
> Second, extracting watertight UV-textured meshes provides an invaluable supporting output for marine scientists, allowing metric millimeter-scale area and volume calculations in tools like MeshLab while confirming which optical model yields the most accurate physical geometry."

Question 4: "Why did 3D-UIR and SeaSplat outperform Gaussian Splashing on structural similarity despite Gaussian Splashing having higher PSNR?"

- **Target Response Time:** 28 seconds
- **Key Evidence:** PSNR pixel bias vs. SSIM edge preservation; volumetric over-smoothing.

> **Response:**
> "PSNR is a pixel-level mean-squared-error metric that rewards blurry, smooth approximations in hazy regions, which is why Gaussian Splashing scored 27.93 dB on Kwajalein.
> However, on Tokai Maru under severe turbidity, Gaussian Splashing over-smoothed the scene, dropping its SSIM to 0.61.
> In contrast, 3D-UIR explicitly models homodirectional ray gradients, achieving the highest structural similarity across both scenes—0.87 on Kwajalein and 0.70 on Tokai—which preserves the crisp boundary edges essential for detecting real physical change."

Question 5: "What is the exact status of the AIMS EcoRRAP dataset, and how will your pipeline ingest 1.35 terabytes?"

- **Target Response Time:** 25 seconds
- **Key Evidence:** Staged bash scripts; Globus endpoint; institutional DTA in progress.

> **Response:**
> "The Data Transfer Agreement is currently undergoing standard inter-institutional review between AIMS and QUT legal teams.
> To ensure zero project delay, our QUT Aqua HPC pipeline was completely built and validated on Submerged3D and Custom OSCD. We have established our Globus high-speed transfer endpoint, meaning as soon as clearance is granted, the EcoRRAP data will stream directly into our automated six-stage batch queue on cluster Lustre storage."

Presenter's Pre-Flight Checklist (15 Minutes Before Talk)

1. **Display & Aspect Ratio:**
   - [ ] Verify HDMI/DisplayPort output is set to **16:9 widescreen (1920×1080)**.
   - [ ] Confirm PowerPoint (`ENN595_2A_Presentation_Slides.pptx` or `.html`) is running in Presentation Mode.
2. **Media Playback Verification:**
   - [ ] Check that orbiting GIFs loop smoothly on **Slide 5, Slide 11, Slide 12, and Slide 13**.
   - [ ] Ensure high-contrast mesh images render clearly on **Slide 12**.
3. **Pacing Anchor Reminders:**
   - [ ] **Slide 1:** Start slow. Establish authority. Breathe.
   - [ ] **Slide 7 (CONOPS):** You should be at exactly **4:00**. If you are at 3:45, you are perfect. If you are past 4:15, pick up the pace slightly.
   - [ ] **Slide 10 (15/15 Pass):** Show enthusiasm—this is your primary technical achievement!
   - [ ] **Slide 14 (Roadmap):** Finish before 8:00. Never rush the final sentence. Leave room for a crisp "Thank you."
4. **Physical Demeanour:**
   - [ ] Stand upright, open posture, avoid turning your back to look at the screen. Use your printed script or laptop presenter view.
   - [ ] Make eye contact with each member of the examination panel.